

Ahmet Taha Albayrak

ahmetahalbayrak@gmail.com | LinkedIn | GitHub | Personal Website

PROFESSIONAL SUMMARY

AI Engineer with 8+ years of experience designing and delivering production machine-learning systems across computer vision, multimodal AI, agentic workflows, MLOps, and edge inference. Experienced in taking systems from architecture and optimization through deployment and integration, including real-time inference at 30 FPS and data pipelines handling more than 1B image records.

EDUCATION

Middle East Technical University

Graduate studies in Computer Engineering; CGPA: 3.34/4.00

Ankara, Türkiye
Sep 2021 - Sep 2023; on leave since Sep 2023

Bilkent University

B.Sc. in Computer Engineering; CGPA: 3.32/4.00

Ankara, Türkiye
Sep 2013 - Jun 2018

WORK EXPERIENCE

NuveMind Information Technologies

AI Engineer, Co-Founder

Ankara, Türkiye
Mar 2025 - Present

- Multimodal AI Systems:** Architect multimodal pipelines integrating image-processing, language-model, and product-service components for production use.
- Agentic Workflows:** Define tool interfaces and orchestration patterns for agents executing multi-step workflows through external services and APIs.
- System Integration:** Coordinate implementation across AI and software teams, establishing reliable integration and delivery practices.

Pavo Group

Senior Machine Learning Engineer

Ankara, Türkiye
Nov 2021 - Jan 2025

- Multi-Task Learning:** Developed architectures for 2D/3D pose estimation and object detection, improving accuracy by 30% while reducing computational cost by 40%.
- Edge Inference:** Optimized real-time machine-learning inference to 30 FPS on edge devices, delivering a threefold performance improvement.
- IR Sensor Integration:** Integrated IR-camera data into machine-learning pipelines, increasing low-light data-collection capacity by 50%.
- Large-Scale Data Processing:** Designed and operated machine-learning data pipelines handling more than 1B image records.

TÜBİTAK Space Research Institute

Researcher

Ankara, Türkiye
Jul 2020 - Nov 2021

- Distributed Computing:** Investigated volunteer computing for hyperparameter optimization and published the results at the 6th High Performance Computing Conference.
- Microservices:** Containerized services with Docker to improve deployment consistency and scalability.
- Algorithm Development:** Developed machine-learning algorithms using Python, PyTorch, and MXNet.

GCLab Engineering

Computer Engineer

Ankara, Türkiye
Oct 2018 - Jul 2020

- Computer Vision:** Developed vision-based systems and image-processing algorithms with the TÜBİTAK UZAY image-processing team.
- Semantic Search:** Implemented faster and more compact semantic indexing for image and audio data.
- Multimodal Retrieval:** Supported multimodal user queries in internet-facing search systems.

SELECTED PROJECTS

1bitllm

2026

Browser-native playground for quantized Bonsai language models. Designed and implemented client-side ONNX inference with WebGPU, streamed generation through a Web Worker, local model caching, and capability checks without a server-side inference backend.

Technologies: WebGPU, Transformers.js, ONNX, Web Workers

[Live Project](#) | [Source Code](#)

OpenTTD Browser Port

2019

Ported and packaged OpenTTD for browser execution, bringing the C++ simulation engine to a full-screen web canvas through Emscripten and WebAssembly. Integrated the runtime, asset bundle, loading states, and application shell.

Technologies: C++, WebAssembly, Emscripten, Canvas

[Live Project](#) | [Source Code](#)

AWARDS & ACHIEVEMENTS

Bilkent University Senior Project Data Science Award (2017-2018)

National University Admission Examination: Ranked 753rd in the quantitative track among approximately 1.8 million candidates.

Enderun Science High School Salutatorian

PUBLICATION

A.T. Albayrak, İ. Atıl, “Hyper Parameter Optimization for Deep Learning using Volunteer Computing,” 6th High Performance Computing Conference, 2020.

SKILLS

Programming: Python, C++, C, MATLAB, SQL

Machine Learning: PyTorch, TensorFlow, MXNet, NumPy, CUDA

Infrastructure: Docker, Git, MLOps, Microservices

Specialization: Computer Vision, Multimodal AI, Agentic AI, Edge AI, Semantic Search

Languages: Turkish (Native), English (Professional)